

Problem 16.11

Which of the following are true?

(i) $\ddot{s}_{\overline{9}|} + 1 = s_{\overline{10}|}$

(ii) $a_{\overline{10}|} - a_{\overline{3}|} = v^2 a_{\overline{7}|}$

(iii) $a_{\overline{5}|}(1+i) - 1 = a_{\overline{4}|}$

Solution

We check each statement separately.

Statement (i)

Using

$$\ddot{s}_{\overline{n}|} = (1+i)s_{\overline{n}|},$$

we have

$$\begin{aligned}\ddot{s}_{\overline{9}|} + 1 &= (1+i) \left(\frac{(1+i)^9 - 1}{i} \right) + 1 \\ &= \frac{(1+i)^{10} - (1+i)}{i} + 1 \\ &= \frac{(1+i)^{10} - (1+i) + i}{i} \\ &= \frac{(1+i)^{10} - 1}{i} \\ &= s_{\overline{10}|}.\end{aligned}$$

So statement (i) is true.

Statement (ii)

Using

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

we get

$$\begin{aligned}
a_{\overline{10}} - a_{\overline{3}} &= \frac{1 - v^{10}}{i} - \frac{1 - v^3}{i} \\
&= \frac{v^3 - v^{10}}{i} \\
&= v^3 \left(\frac{1 - v^7}{i} \right) \\
&= v^3 a_{\overline{7}}.
\end{aligned}$$

This is not equal to $v^2 a_{\overline{7}}$. Therefore, statement (ii) is false.

Statement (iii)

Using

$$a_{\overline{5}} = \frac{1 - v^5}{i},$$

we have

$$\begin{aligned}
a_{\overline{5}}(1 + i) - 1 &= \left(\frac{1 - v^5}{i} \right) (1 + i) - 1 \\
&= \frac{(1 + i) - (1 + i)v^5}{i} - 1 \\
&= \frac{(1 + i) - v^4 - i}{i} \\
&= \frac{1 - v^4}{i} \\
&= a_{\overline{4}}.
\end{aligned}$$

So statement (iii) is true.

Hence, the true statements are

$$\boxed{\text{(i) and (iii)}}.$$